Abstract — ExpenseGuard: Desktop-Based Expense Management System Using Java Swing, JDBC, and MySQL

ExpenseGuard is a desktop-based expense management application developed using Java Swing for the graphical user interface, JDBC for database connectivity, and MySQL for backend data storage. The primary objective of this project is to provide users with a simple and efficient platform to record, manage, update, view, and delete their daily expenses through an interactive menu-driven interface.

The system enables users to store expense details such as title, amount, category, and date in a structured database format. It supports essential CRUD (Create, Read, Update, Delete) operations that allow users to maintain accurate financial records. The application ensures reliable communication between the front-end interface and the database using JDBC connectivity, enabling real-time data processing and retrieval.

The project is designed with modular architecture, improving code readability, maintainability, and scalability. The use of Java Swing provides a responsive desktop environment, while MySQL ensures secure and organized data storage. ExpenseGuard serves as a practical solution for personal expense tracking and demonstrates effective integration of GUI development with database management in Java.

This project highlights the implementation of object-oriented programming concepts, event-driven programming, and

database operations, making it suitable as an academic mini-project as well as a foundation for future enhancements such as authentication systems, report generation, and graphical expense analytics.